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Towards Investigating the Role of Artificial Intelligence in Neurological Rehabilitation

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Abstract

Neurological rehabilitation increasingly relies on technology-assisted interventions, with Virtual Reality Rehabilitation Systems (VRRS) demonstrating clinical efficacy across conditions including stroke, traumatic brain injury, and Parkinson's disease. However, the high cost and infrastructure requirements of these systems limit their accessibility. This thesis investigates the feasibility of augmenting VRRS platforms with markerless motion capture, using computer vision to extract clinically relevant movement data from standard video recordings without requiring additional specialised sensors.

Two complementary tracking approaches were implemented and evaluated: MediaPipe pose estimation for anatomical landmark detection and joint angle computation, and colour-based tracking of the blue sphere keywand used in VRRS exercises. Data were collected from 50 participants, comprising 46 healthy volunteers and 4 patients with Parkinson's disease, in 51 recording sessions.

Technical evaluation showed that both methods process frames faster than the 30 frames per second of typical webcam video on standard consumer hardware: ball tracking was substantially faster, while MediaPipe provided the anatomical landmarks needed for upper-limb joint angles. The analysis also clarified the scope of spatial calibration: converting pixels to millimetres using the known diameter of the keywand sphere is relevant to measures derived from the keywand trajectory, such as displacement, speed, and tremor amplitude, but does not affect joint angles, which are invariant to uniform scaling of coordinates. Frequency-based tremor analysis proved unsuitable for the active, goal-directed movements of VRRS exercises, as voluntary movement masks any underlying tremor. Given the small patient group, no clinical conclusions were drawn about

differences between patients and healthy volunteers.

The study establishes that meaningful movement data can be extracted from VRRS sessions using readily available computer vision tools, providing a foundation for larger validation studies correlating video-based features with established clinical measures.

Keywords: Virtual reality rehabilitation, markerless motion capture, MediaPipe, Parkinson's disease, pose estimation, computer vision

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Chapter 1

Introduction

1.1 Background and Motivation

Neurological disorders affect millions of people worldwide, often leaving them with impaired motor function that demands long-term rehabilitation. Conditions like Parkinson's disease, stroke, and spinal cord injury require patients to engage in repetitive exercises over months or years, with progress tracked by clinicians who assess movement quality, range of motion, and symptom severity. The challenge is straightforward: rehabilitation works, but it requires consistent monitoring and feedback that most healthcare systems struggle to provide at scale.

Artificial intelligence and machine learning have started changing this picture. A recent systematic review covering literature from 2014 to 2024 found that AI applications in neurorehabilitation span predictive analytics, robotic therapy systems, brain-computer interfaces, and remote patient monitoring [1]. These technologies promise to extend the reach of rehabilitation beyond clinic walls, enabling patients to train at home while clinicians receive objective data on their progress. The review also identified critical gaps: large-scale validation studies remain scarce, ethical frameworks are underdeveloped, and, most importantly, access to these technologies is limited by cost and infrastructure requirements.

Many AI-powered rehabilitation tools that do exist require expensive hardware, specialised sensors, or proprietary systems that only well-funded clinics can afford. Patients in smaller facilities or developing regions often have no access to these advances. The technology exists, but the delivery mechanism remains a bottleneck.

1.2 The Problem with Current Systems

Consider the Virtual Reality Rehabilitation System (VRRS), developed by Khymeia and widely used in neurological and neuropsychological rehabilitation across Europe. A narrative review examining 30 studies found that VRRS-based neuromotor training often outperforms conventional treatment alone [2]. The system uses virtual reality environments to engage patients in therapeutic exercises while tracking their movements and providing real-time feedback. For conditions ranging from stroke to traumatic brain injury, VRRS has demonstrated measurable benefits.

But there is a catch. The VRRS platform costs upwards of 5,000 euros, requires dedicated space, and demands technical expertise to operate. The review authors explicitly called for cost-effectiveness analysis in future studies, recognising that the field lacks evidence on whether these benefits justify the investment. For a small rehabilitation centre, a rural hospital, or a home-based care programme, systems like VRRS remain out of reach. The clinical value is clear; the accessibility is not.

This creates a gap worth addressing. Effective rehabilitation technology exists, but it is concentrated in facilities that can afford it. Patients who could benefit from objective movement tracking and AI-assisted assessment instead receive subjective evaluations during infrequent clinic visits. The question becomes: can we achieve similar monitoring capabilities without the specialised hardware?

1.3 The Rise of Markerless Motion Capture

A systematic review of 65 studies examined markerless motion capture technology for clinical measurement in rehabilitation [3]. Unlike traditional motion capture systems, which require reflective markers attached to the body and multiple calibrated cameras, markerless approaches use computer vision algorithms to estimate body pose directly from standard video. Systems based on OpenPose, MediaPipe, and similar frameworks can track joint positions and calculate angles using nothing more than a webcam.

The review found that 14 of these studies focused specifically on Parkinson's disease, using extracted features like bradykinesia characteristics and tremor patterns for machine learning classification. Another 10 studies measured gait features for various conditions, while 4 used whole-body motion analysis for disease classification. What emerges from this body of work is a clear pattern: markerless motion capture is no longer experimental. It has reached the point where researchers across multiple clinical domains are actively validating it against established measurement methods.

The appeal is obvious. A patient performing rehabilitation exercises at home could be monitored through their laptop camera. A clinic without a budget for specialised equipment could still collect objective movement data. The barrier to entry drops from thousands of euros to essentially zero additional cost, since the required hardware already exists in most settings.

1.4 MediaPipe as a Validated Tool

Among the available markerless pose estimation frameworks, Google's MediaPipe has gained particular traction for rehabilitation applications. A 2023 study validated MediaPipe for measuring range of motion in rehabilitation exercises by comparing it against a universal goniometer and a digital angle ruler [4]. The researchers collected data from approximately 50 healthy volunteers with assistance from physical therapists, measuring joint angles during standardised movements.

The results were encouraging. Intraclass correlation coefficient analysis demonstrated

excellent reliability, and Bland-Altman plots confirmed that MediaPipe measurements fell within clinically acceptable limits of agreement. The authors went further, developing a telerehabilitation website that incorporated MediaPipe for remote patient monitoring, demonstrating that the technology could be deployed in practical settings rather than just laboratory conditions.

This validation matters because it addresses the scepticism that often surrounds computer vision approaches in clinical contexts. Clinicians rightfully demand evidence that a measurement tool is accurate before they trust it for patient care. The Latreche study provides exactly that evidence for MediaPipe-based joint angle measurement, using a methodology that mirrors standard validation practices for any clinical instrument.

1.5 Computer Vision for Parkinson's Assessment

The connection between computer vision and Parkinson's disease assessment has been explored most thoroughly in the context of bradykinesia, the slowness of movement that characterises the condition. A study published in npj Parkinson's Disease used video recordings from four clinical sites and one research site across two continents to build computer vision models for bradykinesia severity estimation [5].

The system achieved an intraclass correlation of 0.74 with clinician ratings for composite bradykinesia scores. For 84% of assessments, the disagreement between model and clinician fell below the minimal clinically important difference threshold. In practical terms, a computer vision system using standard video without specialised sensors can assess bradykinesia with accuracy comparable to trained clinicians. The multi-site nature of the dataset strengthens confidence that these results generalise beyond a single laboratory setting.

This work demonstrates that the markerless approach is not limited to simple joint angle measurement; it can also capture the subtle movement characteristics that neurologists use to assess disease severity. For Parkinson's patients, who require regular monitoring to track disease progression and medication response, camera-based assess-

ment could enable more frequent evaluation without requiring clinic visits.

1.6 Proposed Approach

This thesis investigates the application of MediaPipe-based computer vision for movement analysis in neurological rehabilitation, with a focus on developing an accessible, low-cost method to augment systems like VRRS. The work centres on tracking upper limb movements during rehabilitation exercises, extracting joint angles for the shoulder, elbow, and wrist, and analysing movement patterns that may indicate motor symptoms.

The system processes video recordings of patients performing exercises with the VRRS blue sphere keywand, a handheld device used in virtual reality rehabilitation games. By combining MediaPipe pose estimation with colour-based object tracking, the system captures both body posture and hand position throughout the exercise. Spatial calibration converts pixel measurements to millimetres using the known sphere diameter, enabling quantitative analysis of movement amplitude and speed.

Data was collected from 50 participants, including 46 healthy volunteers and 4 patients with Parkinson's disease. The analysis pipeline extracts features relevant to motor symptoms, including bradykinesia, assessed through segment-based analysis of movement speed and amplitude, and movement variability that may indicate dyskinesia. Frequency analysis was also implemented to investigate tremor detection, though, as discussed in later chapters, this proved unsuitable for the active movement context of VRRS exercises.

The contribution is primarily technical rather than clinical. The goal is not to replace clinical diagnosis but to demonstrate that meaningful movement data can be captured and analysed using accessible technology. If successful, this approach could supplement existing rehabilitation systems, enabling more frequent monitoring and objective progress tracking without requiring additional hardware investment.

1.7 Scope and Limitations

Several boundaries define what this thesis does and does not attempt. The work focuses on upper limb movement analysis during a specific type of rehabilitation exercise; it does not address gait analysis, balance assessment, or other movement domains that would require different tracking approaches. The participant pool, while sufficient for technical validation, is too small for robust clinical conclusions about diagnostic accuracy.

The symptom analysis components carry important caveats. In particular, tremor detection through frequency analysis requires the patient to maintain a rest position, while the exercises analysed here involve continuous voluntary movement. This limitation and its implications are discussed in detail in Sections 3.7.3 and 4.5.

The system also inherits the limitations of its underlying technologies. MediaPipe pose estimation can fail when limbs are occluded, when lighting conditions are poor, or when the subject moves outside the camera frame. The blue sphere tracking depends on colour detection that may be affected by environmental lighting. These practical constraints affect the reliability of data collection in real-world settings.

1.8 Thesis Structure

The remainder of this document is organised as follows:

Chapter 2 presents the state of the art, reviewing virtual reality rehabilitation systems, markerless motion capture technology, and computer vision approaches to movement disorder assessment.

Chapter 3 describes the methodology in detail, including the system architecture, signal processing pipeline, calibration procedures, and feature extraction algorithms.

Chapter 4 presents the results: data collection and detection rates, computational performance, position tracking and the role of spatial calibration, frequency analysis, data quality, and a comparison of the two tracking methods, followed by a discussion of the findings and their limitations.

Chapter 5 concludes with a summary of the approach and results, the contributions of this work, its limitations, and directions for future work.

Chapter 2

State of the Art

2.1 Virtual Reality in Neurological Rehabilitation

2.1.1 Historical Development

The application of virtual reality to rehabilitation medicine began gaining traction in the mid-1990s, initially as an experimental approach to make repetitive therapy more engaging. Early systems were crude by modern standards, offering limited immersion and requiring substantial technical expertise to operate. The fundamental premise, however, proved sound: patients who found rehabilitation exercises tedious were more likely to adhere to therapy when those exercises were embedded within interactive virtual environments.

By the early 2000s, commercial systems began appearing in specialized rehabilitation centers. These platforms combined motion tracking hardware with software that transformed physical movements into virtual interactions, allowing patients to practice reaching, grasping, and other motor tasks within game-like scenarios. The therapeutic rationale drew from motor learning theory, particularly the principle that task-specific, repetitive practice with feedback promotes neural plasticity and functional recovery [6].

The distinction between immersive and non-immersive virtual reality became important during this period. Immersive systems use head-mounted displays to replace the

user's visual field entirely with computer-generated imagery, creating a sense of presence within the virtual environment. Non-immersive systems present virtual content on screens while the user remains visually anchored to the physical world. Both approaches found applications in rehabilitation, with the choice depending on the specific therapeutic goals and patient tolerance [2].

2.1.2 The VRRS Platform

Among the systems that emerged from this development, the Virtual Reality Rehabilitation System (VRRS) developed by Khymeia represents one of the most clinically established platforms in European neurological rehabilitation. As introduced in Section 1.2, VRRS provides both neuromotor and cognitive rehabilitation modules with capabilities for immersive and non-immersive operation [2].

The technical architecture merits closer examination. The system centers on a computer connected to a six-degree-of-freedom electromagnetic motion tracking system, typically the Polhemus G4, paired with a high-resolution display. Patients interact with virtual scenarios using a sensorized glove or a handheld tracking device, often a blue sphere keywand that the system tracks in three-dimensional space. The motion tracking enables real-time mapping of physical movements to virtual object manipulation, creating the feedback loop essential for motor learning [7].

A narrative review examining 30 studies using VRRS found applications spanning multiple neurological conditions [2]. Neuromotor rehabilitation was the focus of 13 studies, cognitive rehabilitation accounted for 11, and 6 addressed both domains. The conditions treated included stroke, traumatic brain injury, spinal cord injury, multiple sclerosis, and Parkinson's disease. Study designs varied substantially, including 15 randomized controlled trials alongside case studies and observational work.

Clinical outcomes generally favored VRRS-based intervention, though the review authors noted important limitations. Most studies compared VRRS to conventional therapy rather than to other VR systems, making it difficult to determine whether benefits stemmed from the specific platform or from VR rehabilitation more generally. The

heterogeneity of study designs complicated meta-analytic synthesis, and the review explicitly called for cost-effectiveness analysis to justify the substantial investment these systems require [2].

2.1.3 Limitations of Current Systems

Beyond the cost constraints discussed in Section 1.2, the practical limitations of systems like VRRS extend to their operational requirements. The electromagnetic tracking systems require dedicated space free from metallic interference. Operating the software and interpreting the motion data demands trained personnel, adding ongoing staffing costs to the initial capital investment.

These constraints become particularly problematic when considering the scale of neurological rehabilitation needs. Parkinson's disease alone affects over 10 million people worldwide, with prevalence increasing as populations age. Stroke survivors number in the tens of millions globally, many requiring sustained rehabilitation over months or years. The mismatch between the infrastructure demands of specialized systems and the distributed nature of rehabilitation needs creates a clear accessibility gap.

The COVID-19 pandemic highlighted this gap acutely. When in-person clinic visits became difficult or impossible, patients with neurological conditions faced interruptions in their rehabilitation programs. Systems requiring specialized hardware and trained operators could not easily transition to remote delivery, underscoring the need for rehabilitation technologies that work in home environments with consumer-grade equipment [1].

2.2 Motion Capture Technology

2.2.1 Marker-Based Systems

The gold standard for biomechanical motion analysis has long been marker-based optical motion capture. Systems such as Vicon and Qualisys use multiple infrared cameras

to track retroreflective markers attached to anatomical landmarks on the body. By triangulating marker positions across camera views, these systems achieve submillimeter accuracy in three-dimensional position estimation [8].

The precision of marker-based systems comes at substantial cost, both financial and practical. A typical installation requires eight or more specialized cameras, professional calibration, and a controlled laboratory environment. Marker placement follows standardized protocols that demand anatomical knowledge and consistent technique. Each data collection session involves applying markers to the subject, a process that is time-consuming for a full-body setup and introduces variability between sessions when markers are removed and reattached.

For clinical applications, additional limitations emerge. Patients with neurological conditions may find marker attachment uncomfortable or distressing. Markers can shift during movement, particularly over soft tissue, introducing artifacts into the captured data. The laboratory setting itself may alter natural movement patterns, as patients behave differently under observation in unfamiliar environments than they would at home [8].

2.2.2 The Emergence of Markerless Approaches

Markerless motion capture eliminates the need for physical markers by using computer vision algorithms to estimate body pose directly from video. The approach relies on machine learning models trained to recognize human body configurations and predict the locations of anatomical landmarks from pixel data. When successful, markerless methods preserve the quantitative capability of motion analysis while removing the practical barriers associated with marker attachment.

The systematic review by Lam et al. introduced in Section 1.3 provides the most comprehensive picture of the field's current state [3]. Beyond the Parkinson's-focused studies noted earlier, the review documented applications spanning gait analysis, balance assessment, and upper limb function evaluation across conditions including stroke, cerebral palsy, and orthopedic injuries. The breadth of clinical domains adopting markerless

methods reflects growing confidence in their practical utility.

The shift from marker-based to markerless methods represents more than a technical substitution. By reducing equipment requirements to a standard camera, markerless approaches enable motion analysis in settings where laboratory systems could never operate. Home environments, outpatient clinics, and even public spaces become viable data collection locations. The potential for remote monitoring through consumer devices opens possibilities for continuous assessment that would be impractical with any marker-based approach.

2.2.3 Validation Against Gold Standards

The clinical utility of markerless motion capture depends on demonstrated accuracy relative to established methods. Several studies have compared markerless systems to marker-based gold standards, with results that vary depending on the specific task and measurement.

Multi-camera markerless systems using pose estimation algorithms report average joint center location differences between 10 and 50 mm compared to marker-based references [8]. For temporospatial gait parameters, differences of approximately 15 mm have been observed. These error magnitudes may be acceptable for some clinical applications while proving insufficient for others, underscoring the importance of matching the technology to the measurement requirement.

Single-camera systems, which offer the greatest practical advantages, typically show larger errors than multi-camera setups but remain useful for many applications. A validation study comparing a stereoscopic depth camera with Vicon for center of mass estimation during quiet standing and sway tasks found good agreement in the sway tasks, although the limits of agreement were wider than in quiet standing [9]. The authors concluded that single-camera systems show promise for clinical applications where the trade-off between accuracy and accessibility favors the latter.

A recent validation study compared the Theia3D markerless system against Vicon during quiet stance, using principal component analysis to assess postural movements

[10]. The first two principal movements showed near-perfect correlation between systems, with coefficients of 99.8% and 99.0%. Higher-order components showed declining agreement, suggesting that markerless methods capture gross movement patterns well but may miss subtle details present in marker-based data.

2.3 Deep Learning Pose Estimation

2.3.1 From Classical Methods to Neural Networks

Human pose estimation underwent a fundamental transformation with the application of deep learning beginning around 2014. Classical approaches relied on hand-crafted features and graphical models to infer body configuration from images. These methods achieved reasonable accuracy under controlled conditions but struggled with the variability of real-world imagery, including diverse backgrounds, lighting conditions, and body appearances [11].

DeepPose, introduced by Toshev and Szegedy in 2014, applied convolutional neural networks to pose estimation by framing it as a regression problem toward body joint coordinates [12]. The approach achieved substantial improvements over classical methods and established the paradigm that would drive subsequent development: train neural networks on large annotated datasets to learn the mapping from pixels to pose.

The years following DeepPose saw rapid progress. Stacked hourglass networks introduced multi-scale processing for improved accuracy. Part affinity fields enabled pose estimation for multiple people simultaneously. Attention mechanisms and transformer architectures further refined the ability to model relationships between body parts. Each advance expanded the conditions under which reliable pose estimation could be achieved [13].

2.3.2 OpenPose

OpenPose, released by Cao and colleagues in 2017, became the first widely adopted open-source framework for real-time multi-person pose estimation [13]. The system uses a multi-stage convolutional neural network architecture that simultaneously predicts confidence maps for joint presence and part affinity fields encoding connections between body parts. By processing these outputs together, OpenPose can detect and track multiple people in a scene without requiring person detection as a preprocessing step.

The framework detects 18 body keypoints in its default configuration, covering the major joints of the head, torso, and limbs. Variants extend this to include hand and face keypoints, enabling more detailed gesture and expression analysis. Processing speed depends on hardware but can approach real-time on consumer graphics processors.

OpenPose found immediate application in rehabilitation research. Its ability to extract motion data from standard video without specialized equipment aligned well with the practical constraints of clinical settings. Studies applied OpenPose to gait analysis, balance assessment, and upper limb function measurement, establishing baseline performance for the emerging field of video-based clinical motion analysis [5].

2.3.3 MediaPipe

MediaPipe, developed by Google and released in 2019, built upon the foundation established by earlier pose estimation frameworks while emphasizing mobile and edge deployment. The BlazePose model within MediaPipe achieves real-time performance on smartphones and consumer computers through an architecture optimized for efficiency [14].

MediaPipe Pose tracks 33 landmarks on the human body, more than OpenPose's default 18. The expanded set includes additional points on the hands, feet, and face that enable finer-grained movement analysis. The framework operates in a two-stage pipeline: an initial detection phase locates the person in the frame, followed by a tracking phase that estimates landmark positions with reference to the previous frame. By limiting

detection to occasional frames, the system maintains high frame rates during continuous tracking.

The practical advantages of MediaPipe for rehabilitation applications are substantial. No specialized hardware is required beyond a camera, which can be a smartphone, laptop webcam, or inexpensive USB device. The software is open-source and well-documented, lowering barriers to implementation. Processing can occur locally without transmitting video data, addressing privacy concerns in healthcare contexts.

The Latreche et al. validation discussed in Section 1.4 confirmed that MediaPipe achieves sufficient accuracy for rehabilitation joint angle measurement [4]. Their development of a functional telerehabilitation website further demonstrated that the technology translates from laboratory validation to practical clinical deployment.

2.3.4 Comparative Analysis

The proliferation of pose estimation frameworks creates a need for systematic comparison. A comprehensive review examining machine learning pose estimation models used in human movement analysis identified key differences in architecture, landmark count, and validated applications [15].

Table 2.1 summarizes the major pose estimation frameworks relevant to rehabilitation applications.

Table 2.1: Comparison of pose estimation frameworks for rehabilitation applications

Framework	**Landmarks**	**Real-time**	**Mobile**	**Open Source**
OpenPose	18/25 body	Yes	Limited	Yes
MediaPipe	33 body	Yes	Yes	Yes
BlazePose	33 body	Yes	Yes	Yes
AlphaPose	17 body	Yes	Limited	Yes
MoveNet	17 body	Yes	Yes	Yes
HRNet	17 body	No	No	Yes
DeepLabCut	Custom	Variable	No	Yes

The choice among frameworks depends on the specific application requirements. OpenPose remains widely used for research applications where its extensive validation

provides confidence. MediaPipe has become dominant for mobile and embedded applications where computational efficiency matters. DeepLabCut offers flexibility for custom landmark definitions but requires training on application-specific data. HRNet achieves higher accuracy at the cost of computational speed, making it suitable for offline analysis of recorded video.

2.4 Computer Vision for Parkinson's Disease Assessment

2.4.1 Motor Symptoms and Clinical Rating

Parkinson's disease manifests through a constellation of motor symptoms, with bradykinesia, tremor, and rigidity serving as cardinal features for diagnosis. The Movement Disorder Society revision of the Unified Parkinson's Disease Rating Scale (MDS-UPDRS) provides a standardized framework for assessing these symptoms through a series of motor tasks rated on a 0-4 scale by a clinician [16].

Part III of the MDS-UPDRS covers the motor examination, including items 3.4 through 3.8 that specifically assess bradykinesia through finger tapping, hand movements, pronation-supination, toe tapping, and leg agility. Item 3.15 assesses postural tremor, while item 3.17 addresses rest tremor. Each item requires the patient to perform a specific action while the examiner rates the observed impairment based on speed, amplitude, rhythm, and fatigability [16].

While the MDS-UPDRS provides a common language for clinical assessment, the rating process remains subjective. Inter-rater reliability, though improved over the original UPDRS, still shows meaningful variability. Studies comparing ratings from different movement disorder specialists on the same video recordings reveal disagreement that affects clinical decision-making. This subjectivity motivates the search for objective, quantitative measurement methods [17].

2.4.2 Tremor Frequency Analysis

Tremor assessment through computer vision presents a well-defined technical challenge: extract oscillatory movement signals from video and determine their frequency characteristics. The Parkinson's resting tremor typically occurs at 4-6 Hz, distinguishable from essential tremor at 6-8 Hz and physiological tremor at higher frequencies. Accurate frequency estimation can support differential diagnosis and track treatment response.

A study validating smartphone video for tremor frequency measurement compared computer vision analysis against accelerometer recordings in patients with Parkinson's disease, essential tremor, and functional tremor [18]. Using optical flow to measure pixel movement, the researchers achieved excellent agreement with accelerometer-derived frequencies. Bland-Altman analysis showed 95% limits of agreement of -0.38 Hz to +0.35 Hz, with 97% of measurements differing by less than 0.5 Hz from the accelerometer reference.

More recent work has validated MediaPipe for video-based tremor analysis against motion capture [19]. In two clinical cohorts totalling 66 patients with essential tremor, MediaPipe-derived tremor measures were compared with marker-based motion capture, wrist-worn accelerometry and clinical scores. The video measures reached an accuracy of 0.21 Hz or better for tremor frequency and up to 2.6 mm for amplitude, comparable to gold-standard equipment, and correlated with clinical tremor scores (Spearman's $\rho = 0.55$–0.86).

Table 2.2 summarizes validation studies for video-based tremor analysis.

Table 2.2: Validation studies for video-based tremor frequency measurement

Study	**Method**	**n**	**Frequency Error**	**Reference**
Williams 2021	Optical flow	37 videos	-0.38 to $+0.35$ Hz	Accelerometer
Friedrich 2024	MediaPipe	66 patients	≤ 0.21 Hz	OMC, Accel

2.4.3 Bradykinesia Quantification

Bradykinesia assessment through computer vision has received substantial attention due to the symptom's centrality to Parkinson's diagnosis and the well-defined nature of the assessment tasks. The finger tapping item of the MDS-UPDRS (item 3.4) provides an ideal target: patients tap their index finger against their thumb as quickly and widely as possible while the examiner observes speed, amplitude, decrement, and hesitations.

DeepLabCut was applied to smartphone videos of finger tapping, with 22 movement disorder neurologists rating the same recordings [20]. Computer-derived measures of speed, amplitude, and rhythm correlated significantly with clinical ratings: Spearman coefficients reached -0.74 for speed on the Modified Bradykinesia Rating Scale.

The Morinan et al. multi-site study introduced in Section 1.5 expanded this work substantially [5]. By collecting data across four clinical sites and one research site spanning two continents, they demonstrated that video-based bradykinesia assessment generalises beyond individual laboratory settings. The scale of validation (1156 assessments) and the clinical-grade agreement (ICC=0.74) established a benchmark for the field.

Subsequent studies have refined these approaches. FastEval Parkinsonism combined MediaPipe hand tracking with deep learning classification to create an online system for bradykinesia evaluation [21]. Its developers reported acceptable accuracy on validation data, demonstrating the feasibility of deployment beyond research settings.

Table 2.3 presents key validation studies for video-based bradykinesia assessment.

Table 2.3: Validation studies for video-based bradykinesia assessment

Study	**Method**	**n**	**Correlation**	**Task**
Williams 2020	DeepLabCut	133 hands	$\rho = -0.74$ (speed)	Finger tap
Morinan 2023	OpenPose	1156 assess.	ICC=0.74	Multiple

2.4.4 Challenges and Limitations

Despite promising results, several challenges limit the clinical adoption of video-based movement disorder assessment. The distinction between task performance during a clin-

ical assessment and symptom manifestation in daily life remains important. A patient performing finger tapping on command may exhibit different characteristics than the same patient during unstructured activity. Transfer of validation results across contexts cannot be assumed.

Tremor assessment faces a particular limitation when applied during active movement. Clinical tremor scales typically assess tremor at rest or during maintained posture, conditions that maximize the visibility of involuntary oscillation. During active task performance, voluntary movement signals dominate, making it difficult to isolate pathological tremor. This context-dependence becomes directly relevant to the present work, as discussed in Sections 3.7.3 and 4.5.

Technical limitations also constrain performance. Lighting conditions affect detection reliability, with poor illumination reducing pose estimation confidence. Camera angle influences which body parts are visible and how accurately their positions can be estimated. Occlusion by furniture, other people, or the patient's own body parts causes tracking failures that may occur non-randomly with respect to symptom severity.

2.5 Comparison of Motion Capture Approaches

2.5.1 Performance Metrics

Evaluating motion capture systems requires attention to multiple performance dimensions. Accuracy describes how closely estimated positions match true positions, typically quantified as root mean square error or mean absolute error in millimeters. Precision captures the variability of repeated measurements under identical conditions. Temporal resolution, measured in frames per second, determines whether rapid movements can be captured. Detection reliability indicates the proportion of frames in which the system successfully tracks the target.

For clinical applications, the clinically meaningful difference provides context for interpreting accuracy figures. Joint angle errors of 5 degrees may be clinically irrelevant for gross movement assessment while proving problematic for detailed gait analysis. The

appropriate accuracy threshold depends on the specific clinical question rather than on any universal standard.

2.5.2 Comparative Studies

Table 2.4 synthesizes findings from comparative studies of marker-based and markerless motion capture systems.

Table 2.4: Comparison of motion capture approaches for rehabilitation

Characteristic	**Marker-Based**	**Multi-Cam ML**	**Single-Cam ML**
Position accuracy	<1 mm	10-50 mm	30-100 mm
Setup time	20-30 min	5-10 min	<1 min
Equipment cost	$50,000+	$5,000-20,000	$0-500
Portability	Laboratory	Limited	High
Operator expertise	High	Moderate	Low
Real-time capable	Yes	Variable	Yes
Home deployment	No	Difficult	Yes

The trade-offs are clear. Marker-based systems achieve unmatched accuracy but require substantial infrastructure and expertise. Multi-camera markerless systems reduce setup burden while maintaining reasonable accuracy for many applications. Single-camera markerless systems, including smartphone-based approaches, sacrifice accuracy for accessibility but enable entirely new use cases in home monitoring and telemedicine.

2.5.3 Application-Specific Considerations

Different rehabilitation applications impose different requirements on motion capture performance. Gait analysis for surgical planning may demand the submillimeter accuracy that only marker-based systems provide. Gross upper limb tracking for stroke rehabilitation may tolerate the larger errors of single-camera systems while benefiting from their deployability. Tremor frequency estimation, which depends on temporal rather than spatial precision, may be well-suited to video-based methods even when amplitude estimation remains challenging.

The choice of motion capture approach should match the clinical question. For applications where accessibility matters more than precision, markerless methods offer clear advantages. For applications requiring detailed kinematic analysis, the investment in marker-based systems may remain justified. Hybrid approaches that combine markerless tracking for frequent monitoring with occasional marker-based assessment for detailed analysis represent a potential middle ground [8].

2.6 Gaps and Opportunities

2.6.1 The Accessibility Problem

The preceding sections establish two facts. First, effective rehabilitation technology like VRRS exists but remains inaccessible to most patients due to cost and infrastructure requirements. Second, markerless motion capture has matured to the point where it can extract clinically meaningful movement data using only consumer-grade cameras and free software. The gap between these two facts defines the opportunity this thesis addresses.

2.6.2 Integration with Existing Rehabilitation Systems

Rather than replacing specialized platforms, an opportunity exists to augment them. Systems like VRRS already generate video output during rehabilitation sessions. Adding markerless analysis to this existing video stream could extract additional movement information without requiring new sensors. The blue sphere keywand used in VRRS provides a convenient tracking target detectable through colour-based methods, complementing the body pose information available through MediaPipe.

Such a hybrid approach would preserve the therapeutic value of VR-based rehabilitation while adding quantitative movement analysis capabilities. Clinicians could receive objective measures of joint angles, movement speed, and symptom indicators alongside the task performance data already captured by the VR system.

2.6.3 From Laboratory to Clinical Practice

Moving computer vision methods from research validation to clinical deployment requires addressing practical concerns that research studies often defer. Robustness to real-world conditions, including variable lighting, cluttered backgrounds, and patient positioning variations, becomes critical. User interfaces must accommodate clinicians without technical backgrounds. Data management must comply with healthcare privacy regulations.

The technical foundations for video-based movement analysis are increasingly mature. The remaining barriers are primarily translational: demonstrating clinical utility in realistic settings and establishing the evidence base that healthcare systems require for adoption. The following chapter details the methodology developed to address these challenges within the VRRS rehabilitation context.

Chapter 3

Materials and Methods

3.1 Study Design

This study follows an observational cross-sectional design in which healthy volunteers and patients with Parkinson's disease were video-recorded while performing VRRS rehabilitation exercises. The primary objective is to demonstrate that meaningful movement data can be extracted from standard video recordings using markerless motion capture, potentially extending the monitoring capabilities of existing rehabilitation systems without requiring additional specialised equipment.

The study involves two complementary tracking approaches. The first tracks the blue sphere keywand used in VRRS exercises using colour-based detection. The second employs MediaPipe pose estimation to capture body joint positions and compute upper limb angles. Both approaches process video recordings captured during rehabilitation sessions, enabling comparison of their respective strengths and computational requirements.

3.1.1 Participants

Data were collected from 50 participants: 46 healthy volunteers and 4 patients with diagnosed Parkinson's disease. The volunteer group comprised 47 recording sessions from

46 unique individuals, and the patient group comprised 4 sessions from 4 individuals, for a total of 51 recordings across 50 participants. With only four patients, the dataset supports a technical feasibility study rather than a statistical comparison between the two groups.

Note that one volunteer (subject 27VM/30VM) was recorded twice under different identifiers; both recordings were retained in the dataset. All participants performed upper limb exercises using the VRRS system while video was simultaneously recorded for offline analysis.

3.1.2 Ethical Considerations

All participants provided informed consent prior to data collection. Video recordings were processed locally without transmission to external servers, and all stored data were anonymized using alphanumeric identifiers (e.g., 10MVM, 1PM) that contain no personally identifiable information.

3.2 Hardware Configuration

3.2.1 VRRS System

The VRRS platform described in Section 2.1.2 served as the rehabilitation platform for data collection. Patients manipulated the blue sphere keywand, approximately 50 mm in diameter, to interact with virtual scenarios displayed on the system's high-resolution screen. The sphere's known diameter enables spatial calibration (see Section 3.3.5), and its distinctive blue colour facilitates detection through colour-based computer vision methods.

3.2.2 Recording Setup

Sessions were recorded with OBS Studio as screen recordings that included the view of a standard webcam positioned to capture the participant and the interaction space.

Recording parameters were:

- Frame rate: 60 frames per second for most recordings (30 frames per second for some)
- Resolution: 3360 × 2100 pixels for most recordings (1280 × 720 pixels for some)
- Format: MKV container, H.265 (HEVC) or H.264 video
- Duration: variable, typically 8–12 minutes per session

The four patient sessions were recorded by a clinician on a different computer, with a different screen layout. For processing, each frame was downscaled, by default by a factor of 0.6 (to 2016 × 1260 pixels for a 3360 × 2100 recording). All pixel quantities reported in this thesis refer to the downscaled frames.

For processing, each frame was downscaled by a factor of 0.6, to 2016 × 1260 pixels. All pixel quantities reported in this thesis refer to this downscaled resolution.

Environmental conditions such as camera distance, lighting, and background were deliberately not standardised across sessions. This choice reflects the intended deployment context: VRRS exercises are typically carried out with minimal setup guidance, and a system intended to augment such sessions needs to tolerate the same variability. Recordings were therefore collected under a range of naturalistic conditions, including sessions captured under morning daylight and others under artificial lighting in the evening. No specialised lighting or background preparation was required, and the recordings captured rehabilitation sessions as they would occur in standard clinical practice.

Within each session, however, the camera remained in a fixed position and the participant stayed at an approximately constant distance from it. Because camera distance differed between sessions but not within them, the pixel-to-millimetre calibration is estimated separately for each session (Section 3.3.5).

3.3 Software Environment and Ball Tracking

All processing was implemented in Python 3.12 using the libraries listed in Table 3.1.

Table 3.1: Software dependencies and versions

Library	Version	Purpose
OpenCV	4.11.0	Video processing, colour detection, Kalman filtering
MediaPipe	0.10.21	Pose estimation
NumPy	1.26.4	Numerical computation, FFT
SciPy	1.13.1	Interpolation, Butterworth filtering, peak detection
Pandas	2.2.2	CSV export
Matplotlib	3.9.2	Figures

3.3.1 Testbed Configuration

All processing, including the computational benchmarking reported in Section 4.2, was performed on a MacBook Pro (13-inch, 2020) with an Apple M1 chip and 8 GB of unified memory. No GPU acceleration was used; both MediaPipe and the ball tracking pipeline ran on CPU only. Reporting this configuration is necessary because the processing-time and resource-utilisation figures in Chapter 4 are specific to this hardware and are not intended as absolute performance figures for other systems.

3.3.2 Colour-Based Sphere Detection

The sphere is searched for within a rectangular region of interest (ROI) of each down-scaled frame, by default measuring 498×375 pixels and covering the area of the recording in which the keywand moves. All sphere coordinates are expressed in the pixel coordinates of this region. The ROI is converted from BGR (OpenCV's default input format) to the HSV colour space, and a binary mask is obtained by thresholding the three channels:

$$M(x,y) = \begin{cases} 1 & \text{if } 85 \leq H \leq 140,\ S \geq 100,\ V \geq 120 \\ 0 & \text{otherwise} \end{cases} \tag{3.1}$$

where hue is expressed on OpenCV's 0–179 scale and saturation and value on a 0–255 scale. Because HSV separates colour from brightness, the blue of the sphere remains identifiable under moderate changes in illumination, although strong lighting changes and reflective surfaces still reduce detection reliability (Section 4.1.2).

The values given in this chapter are the default settings of the pipeline. Because the lighting and the screen layout differed between recordings, some settings had to be adjusted by hand for individual recordings so that the sphere could be detected. The region of interest was changed for recordings with a different layout, including the four patient sessions, and for some recordings the colour thresholds were changed or the HSV mask was combined with thresholds in the LAB and YCrCb colour spaces. Not all sessions were therefore processed with identical detection settings.

3.3.3 Morphological Processing

The binary mask undergoes two morphological operations with a 5×5 elliptical structuring element K. Closing (dilation followed by erosion) first fills small holes within the detected region, and opening (erosion followed by dilation) then removes small spurious detections:

$$M_{closed} = (M \oplus K) \ominus K, \qquad M_{processed} = (M_{closed} \ominus K) \oplus K \tag{3.2}$$

where $\oplus$ denotes dilation and $\ominus$ denotes erosion.

3.3.4 Contour Analysis and Position Extraction

External contours are extracted from the processed mask. A contour is accepted as a sphere candidate only if its area A lies between 300 and 20,000 pixels, the radius r of its

minimum enclosing circle lies between 8 and 80 pixels, and its circularity

$$C = \frac{4\pi A}{P^2} \tag{3.3}$$

is at least 0.25, where P is the contour perimeter. The centre and radius of the minimum enclosing circle give each candidate's position and apparent size, and its detection confidence is

$$c = \min\left(1,\ 0.5\,C + 0.5\,\frac{A}{10{,}000}\right) \tag{3.4}$$

When several candidates are present, the one closest to the previous sphere position is selected, provided it lies within 300 pixels. Otherwise, the candidate closest to the position predicted by the Kalman filter (Section 3.5.3) is chosen, with the same 300-pixel limit, and if neither rule applies, the candidate with the highest confidence is used. The centre of the selected candidate is then corrected by the Kalman filter and recorded as the sphere position for that frame. Frames without a valid candidate are counted as missed detections and contribute no position sample; the detection rate of a session is the fraction of processed frames with a valid candidate.

3.3.5 Spatial Calibration

To convert pixel measurements to physical units, the known diameter of the blue sphere (50 mm) serves as a calibration reference. The calibration factor k is expressed in pixels per millimetre, so pixel quantities are converted to millimetres by dividing by k. The apparent diameter of the sphere in a frame is taken as the diameter of its minimum enclosing circle, $d_{px} = 2r$.

Calibration begins with a detection window covering the first 100 frames of the recording (approximately 1.7 seconds at 60 frames per second). The window closes as soon as five detections of consistent size are available, that is, when the coefficient of variation of the last five radii falls below 0.2. The initial calibration factor is then

$$k_0 = \frac{2\bar{r}}{50 \text{ mm}} \tag{3.5}$$

where $\bar{r}$ is the mean of these five radii. If no consistent detection is found within the window, calibration starts from the first detection during tracking.

During tracking, every frame in which the sphere is detected provides a new estimate $k_t^{new} = d_{px,t}/50$ mm. Estimates that differ from the current value by more than a factor of three ($k_t^{new}/k_{t-1} > 3$ or $k_t^{new}/k_{t-1} < 0.33$) are rejected as detection errors, for example when the sphere is partially occluded or merged with another blue region. The calibration factor is the median of the most recent accepted estimates:

$$k_t = \text{median}\left\{k_j^{new} : j \in W_t\right\} \tag{3.6}$$

where W_t contains the last 100 accepted estimates, including k_0.

Because the camera was fixed and the participant's distance from it approximately constant within each session (Section 3.2.2), a single scale factor per session is a reasonable approximation. Individual diameter estimates are nevertheless noisy: they vary with detection errors, partial occlusion by the hand, and the keywand's movements towards and away from the camera. The median over 100 estimates limits the influence of this variability, which is quantified in Section 4.3.3. The cost is a formally time-varying rather than strictly constant scale factor. Measures derived from the keywand trajectory are converted with the final value of k_t at the end of the session, whereas MediaPipe joint positions are converted frame by frame (Section 3.5.3).

3.4 MediaPipe Pose Estimation

3.4.1 Framework Overview

MediaPipe Pose provides real-time human pose estimation from monocular video [14]. The framework detects 33 anatomical landmarks covering the full body. For this study, upper body landmarks were the primary focus, specifically those enabling computation

of shoulder, elbow, and wrist joint angles. Five landmarks of the right side of the body are used:

- Landmark 12: Right shoulder
- Landmark 14: Right elbow
- Landmark 16: Right wrist
- Landmark 20: Right index finger
- Landmark 24: Right hip

3.4.2 Detection Pipeline

MediaPipe operates in a two-stage pipeline. The first stage detects the presence and bounding box of a person in the frame. The second stage estimates landmark positions within that bounding region. By tracking landmarks across frames rather than re-detecting in every frame, the system achieves real-time performance. Each processed frame, downscaled as described in Section 3.2.2 and converted from BGR to RGB, is passed to the model in full; unlike the ball tracker, MediaPipe is not restricted to a region of interest.

The implementation initializes the pose detection model with the following parameters:

Listing 3.1: MediaPipe initialization

```
1 import mediapipe as mp
2
3 mp_pose = mp.solutions.pose
4 pose = mp_pose.Pose(
5     static_image_mode=False,
6     model_complexity=1,
7     smooth_landmarks=True,
8     min_detection_confidence=0.5,
```

```
9       min_tracking_confidence=0.5
10  )
```

The `model_complexity` parameter controls the trade-off between speed and accuracy. A value of 1 provides balanced performance suitable for rehabilitation monitoring. Setting `smooth_landmarks=True` applies MediaPipe's built-in temporal filtering to reduce jitter in landmark positions, serving a similar noise-reduction purpose to the Kalman filter used for ball tracking (Section 3.5.3).

3.4.3 Joint Angle Computation

Joint angles are computed from the two-dimensional normalised image coordinates (x, y) of the landmarks using vector geometry. For a joint defined by three landmarks A, B, and C where B is the joint center, the angle θ is:

$$\theta = \arccos\left(\frac{\vec{BA} \cdot \vec{BC}}{|\vec{BA}| \cdot |\vec{BC}|}\right) \tag{3.7}$$

where $\vec{BA} = A - B$ and $\vec{BC} = C - B$ are the vectors from the joint to the adjacent landmarks.

The implementation computes this as:

Listing 3.2: Joint angle calculation

```
1  import numpy as np
2
3  def calculate_angle(a, b, c):
4      """Calculate angle at point b given points a, b, c."""
5      ba = np.array(a) - np.array(b)
6      bc = np.array(c) - np.array(b)
7
8      cosine = np.dot(ba, bc) / (np.linalg.norm(ba) *
9                                 np.linalg.norm(bc))
10     cosine = np.clip(cosine, -1.0, 1.0)
11     angle = np.degrees(np.arccos(cosine))
```

```
12
13      return angle
```

Three joint angles of the right arm are extracted for each frame in which a pose is detected:

- **Shoulder angle**: Formed by hip, shoulder, and elbow landmarks
- **Elbow angle**: Formed by shoulder, elbow, and wrist landmarks
- **Wrist angle**: Formed by elbow, wrist, and index finger landmarks

MediaPipe normalises the x and y coordinates by the frame width and height respectively. Because the frames are not square (16:10), this scaling is not uniform, so the computed values are not exact image-plane angles; they are therefore treated as relative descriptors and are not used for clinical conclusions. In addition to the angles, the pixel positions of the shoulder, elbow, and wrist are recorded and converted to millimetres using the current calibration factor (Section 3.3.5).

3.5 Signal Processing

3.5.1 Temporal Alignment

Each processed frame is timestamped from its index in the video and the frame rate declared in the file:

$$t_i = \frac{i}{f_{video}} \tag{3.8}$$

where i is the frame index in the original video and f_{video} is the declared frame rate (60 fps for most recordings in this study and 30 fps for some). To reduce processing time, the pipeline can analyse only every n-th frame (a configurable frame-skip parameter). Because timestamps are computed from the original frame index, the time axis is unaffected by this setting. The effective sampling rate of the tracking data is therefore f_{video}/n, further reduced by frames in which the sphere is not detected.

3.5.2 Interpolation to Uniform Sampling

The frequency analysis (Section 3.7) requires uniformly sampled data, whereas the sphere positions are irregularly sampled because of skipped frames and missed detections. Before the FFT, the x and y coordinates of the sphere are therefore interpolated onto a uniform time grid with sampling rate

$$f_s = \max\left(20 \text{ Hz},\ f_{native}\right) \tag{3.9}$$

where f_{native} is the mean sampling rate of the detected positions, computed from the intervals between consecutive samples while ignoring gaps longer than 2 seconds. Interpolation uses a cubic spline; if the spline cannot be fitted, linear interpolation is used instead:

$$x(t) = x_i + \frac{(t - t_i)(x_{i+1} - x_i)}{t_{i+1} - t_i} \tag{3.10}$$

for $t_i \leq t < t_{i+1}$, where x_i and x_{i+1} are consecutive measured values at times t_i and t_{i+1}.

The minimum rate of 20 Hz places the Nyquist frequency at 10 Hz, above the tremor band analysed in Section 3.7 (3.5–7.5 Hz). Interpolation provides the uniform time grid required by the FFT, but it cannot add information above half of the native sampling rate, so the frequency range that can actually be analysed is limited by the processing rate and by gaps in detection. Interpolation is used only for the frequency analysis; all other features are computed from the measured samples and their timestamps.

3.5.3 Kalman Filtering

Position estimates from ball tracking can exhibit noise due to detection uncertainty, occlusion, or motion blur. A Kalman filter provides optimal state estimation by combining prediction based on a motion model with noisy observations. MediaPipe handles noise reduction internally through its `smooth_landmarks` parameter and does not require

additional Kalman filtering.

The state vector for ball tracking includes position and velocity in two dimensions:

$$\mathbf{x} = [x, y, \dot{x}, \dot{y}]^T \tag{3.11}$$

The state transition follows a constant velocity model:

$$\mathbf{x}_{k+1} = \mathbf{F}\mathbf{x}_k + \mathbf{w}_k \tag{3.12}$$

where $\mathbf{F}$ is the state transition matrix:

$$\mathbf{F} = \begin{bmatrix} 1 & 0 & \Delta t & 0 \\ 0 & 1 & 0 & \Delta t \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{3.13}$$

with $\Delta t = 1$, so that velocity is expressed in pixels per processed frame, and $\mathbf{w}_k$ represents process noise with covariance $\mathbf{Q} = 0.01\,\mathbf{I}_4$.

Observations provide position measurements:

$$\mathbf{z}_k = \mathbf{H}\mathbf{x}_k + \mathbf{v}_k \tag{3.14}$$

where $\mathbf{H} = [1, 0, 0, 0; 0, 1, 0, 0]$ extracts position from the state and $\mathbf{v}_k$ is measurement noise with covariance $\mathbf{R} = 0.1\,\mathbf{I}_2$.

The Kalman filter recursively updates state estimates through prediction and correction steps, reducing noise while preserving the underlying movement signal. The filter is implemented with OpenCV's `KalmanFilter` class. In frames where the sphere is detected, the corrected estimate is recorded as the sphere position; in frames without a detection, the filter only predicts, and the prediction is used for display and candidate matching but is not recorded.

In the implementation, the filter's state and error covariance start from zero rather than from the first detection. As a result, the filtered positions at the start of each

recording are pulled towards the origin of the region of interest and converge to the measured positions within about ten detections. This start-up transient affects only the first detections of each recording, but it produces an artificial jump at the beginning of the trajectory that inflates the peak speed of the session.

Notably, the filter operates entirely on raw pixel-space positions; the calibration factor k_t described in Section 3.3.5 is not incorporated into the filter's transition or measurement model. For measures derived from the keywand trajectory, k_t is applied once, as a final scalar conversion using its value at the end of the session: tremor amplitude (after FFT analysis, Section 3.7), average and peak speed and segment distances (Sections 3.6.2 and 3.8), and residual amplitude and jerk (Section 3.8.2). MediaPipe joint positions are converted frame by frame with the current value of k_t, while the sphere positions themselves are stored in pixels. This ordering is a close approximation to filtering already-converted millimetre positions directly: because k_t varies only slowly within a session, Kalman filtering in pixel space and then scaling by k_t produces results very close to scaling first and filtering in millimetres. Filtering in pixel space was preferred because the noise covariances $\mathbf{Q}$ and $\mathbf{R}$ are defined in pixel units; converting to millimetres before filtering would have required re-deriving them in the calibrated unit system. This design choice is noted here as a methodological limitation: applying k_t before filtering, rather than after, would be the more principled approach for a system built from the ground up around calibrated units, and is left as a direction for future refinement.

3.6 Feature Extraction

3.6.1 Position-Based Features

From the tracked sphere positions, the following features are computed:

- **Mean position**: Average x and y coordinates over the recording
- **Position variability**: Standard deviation of the x and y coordinates

- **Path length**: Cumulative distance travelled within each movement segment (Section 3.8.3)

Path length is computed as the sum of Euclidean distances between consecutive positions:

$$L = \sum_{i=1}^{n-1} \sqrt{(x_{i+1} - x_i)^2 + (y_{i+1} - y_i)^2} \tag{3.15}$$

Mean position and position variability are computed from the per-session position files (Section 3.10) and are expressed in pixels, whereas segment path lengths are converted to millimetres with the calibration factor (Section 3.3.5).

3.6.2 Speed

Speed is estimated from consecutive sphere positions by forward differences:

$$v_i = \frac{\sqrt{(x_i - x_{i-1})^2 + (y_i - y_{i-1})^2}}{t_i - t_{i-1}} \tag{3.16}$$

Intervals longer than 2 seconds, which correspond to gaps in detection, are assigned zero speed. The average and peak speed of a session are the mean and maximum of v_i, converted from pixels per second to millimetres per second with the calibration factor (Section 3.3.5). Because differentiation amplifies high-frequency noise, Kalman-filtered positions are used as input; the peak value is nevertheless affected by the start-up transient of the filter described in Section 3.5.3. Acceleration is not computed as a separate feature.

3.6.3 Joint Angle Statistics

For each joint angle time series, summary statistics are computed:

These features capture both the typical joint configuration during the exercise and the extent of movement. For the shoulder, elbow, and wrist, the mean and standard deviation of the joint positions are also computed, in millimetres. As explained in Chapter 4, the

Table 3.2: Extracted joint angle features

Feature	**Description**
Mean angle	Average angle over session
Standard deviation	Angle variability
Minimum angle	Smallest angle reached
Maximum angle	Largest angle reached
Range of motion	Maximum minus minimum

joint angle statistics are part of the system output but are not used for comparisons between groups.

3.7 Frequency Analysis

3.7.1 Fast Fourier Transform

To investigate oscillatory components in the movement signals, Fast Fourier Transform (FFT) analysis was applied to the sphere trajectory. The interpolated x and y coordinates (Section 3.5.2) are first centred by subtracting their means and then filtered with a fourth-order Butterworth band-pass filter covering the tremor band (3.5–7.5 Hz, with the upper edge limited to 95% of the Nyquist frequency). The filter is applied forwards and backwards, so that it introduces no phase shift. The two filtered coordinates are combined into a single magnitude signal:

$$s(n) = \sqrt{x_{bp}(n)^2 + y_{bp}(n)^2} \tag{3.17}$$

which is multiplied by a Hann window $w(n)$ to reduce spectral leakage and transformed with the FFT:

$$X(k) = \sum_{n=0}^{N-1} s(n)\, w(n)\, e^{-j2\pi kn/N} \tag{3.18}$$

where N is the number of samples and bin k corresponds to frequency $f_k = kf_s/N$. The amplitude spectrum is normalised as

$$A(f_k) = \frac{2}{N} |X(k)| \tag{3.19}$$

3.7.2 Tremor Band Analysis

Parkinson's tremor typically manifests at 4–6 Hz [22]. To investigate whether tremor signatures could be detected, the power within the 3.5–7.5 Hz band was computed as the sum of the squared spectral amplitudes in that band:

$$P_{tremor} = \sum_{3.5 \leq f_k \leq 7.5} A(f_k)^2 \tag{3.20}$$

This power is also expressed relative to the total power above 0.1 Hz, giving the tremor ratio $R = P_{tremor}/P_{total}$. The tremor amplitude is the root mean square of the filtered signal $s(n)$, converted to millimetres with the calibration factor (Section 3.3.5), and the dominant tremor frequency is the frequency of the largest spectral amplitude within the band. Because the signal is band-pass filtered before the FFT, most of the voluntary movement is removed before the spectrum is computed.

3.7.3 Limitations of Tremor Detection

A fundamental limitation emerged during analysis. The VRRS exercises require active, goal-directed movement, with patients continuously manipulating the keywand to interact with virtual scenarios. This active movement context differs substantially from the resting state in which Parkinson's tremor is clinically assessed.

Resting tremor, by definition, occurs when the limb is supported and the muscles are not voluntarily activated [22]. During active task performance, voluntary movement signals dominate the motion record, making it difficult to isolate involuntary tremor components. In addition, the tremor band can only be resolved when the native sampling rate of the sphere positions exceeds twice its upper edge (15 Hz), because interpolation cannot recover higher frequencies (Section 3.5.2). The frequency analysis was retained

in the methodology for completeness, but its utility for tremor detection in this context proved limited. Results are presented in Section 4.5.

Detection of rest tremor during VRRS sessions would require either dedicated rest periods where patients pause movement, or inertial measurement unit (IMU) sensors capable of isolating tremor from voluntary motion, neither of which was implemented in this study.

3.8 Symptom Analysis Approach

3.8.1 Bradykinesia Indicators

Bradykinesia, the slowness of movement characteristic of Parkinson's disease, can manifest through reduced movement speed and amplitude, and through a progressive decrease in amplitude over repeated movements (the sequence effect). The following indicators are computed from the sphere trajectory:

- Average and peak speed of the session (Section 3.6.2)
- Peak speed, path length, and duration of each movement segment (Section 3.8.3)
- Speed-to-distance ratio: the peak speed of a segment divided by its path length, averaged over segments
- Amplitude decrement: the relative reduction in mean segment path length between the first and second halves of the session, $D = (\bar{L}_{first} - \bar{L}_{second})/\bar{L}_{first}$, computed when at least four segments are available
- Sub-movements: the number of local speed maxima after the peak of each segment with a prominence above 30% of the mean speed in that phase, averaged over segments, as an indicator of corrective movements

Joint range of motion is also available from the joint angle statistics (Section 3.6.3). These indicators are part of the system output; given the size of the patient group, they were not tested for differences between groups.

3.8.2 Dyskinesia Indicators

Dyskinesia refers to involuntary, excessive movements that can occur in Parkinson's disease, particularly as a side effect of medication. Potential indicators include:

- Increased movement variability beyond task requirements
- Residual motion during intended rest periods
- Irregular trajectory patterns during reaching

The implementation addresses the first and third of these through a residual-based analysis. A smooth trajectory, representing the intended voluntary movement, is obtained by filtering the sphere positions with a third-order Butterworth low-pass filter with a cut-off of 2 Hz (lowered to 40% of the sampling rate if necessary), applied forwards and backwards. The residual is the deviation of the measured positions from this trajectory:

$$r_i = x_i - \hat{x}_i \tag{3.21}$$

where $\hat{x}_i$ is the smoothed position and r_i is the residual, computed for both coordinates. The filter treats the samples as uniformly spaced at their mean sampling rate, so gaps in detection slightly distort the smoothed trajectory. Three indicators are derived from the residual: its root mean square amplitude; a jerk score, which in the implementation is the mean magnitude of the second time derivative of the residual, estimated by finite differences; and an irregularity index, the number of abrupt direction changes in the residual (turns sharper than 60 degrees between consecutive steps longer than 0.5 pixels) divided by the number of samples. The amplitude and the jerk score are

converted with the calibration factor (Section 3.3.5). Elevated residual amplitude may indicate involuntary movement superimposed on voluntary control.

3.8.3 Segment-Based Analysis

Movement recordings are divided into segments corresponding to individual movements, using the speed signal of Section 3.6.2. A segment starts when the speed rises above a threshold and ends when it falls below it again. The threshold is the 25th percentile of the session's speeds, with a minimum of 5 mm/s, so that the slowest quarter of samples is treated as rest. Segments spanning fewer than three samples are discarded to avoid fragmentation from noise. For each segment, peak speed, path length, and duration are computed independently, enabling analysis of movement-to-movement variability within a session and of changes over the session (Section 3.8.1).

3.8.4 Heuristic Screening Output

In addition to the indicators above, the implementation combines them into rule-based severity labels (healthy, mild, moderate, or severe) for tremor, bradykinesia, and dyskinesia, and into a composite score for each session. The thresholds behind these labels and scores were set heuristically and have not been validated against clinical ratings, so they are not used as findings in this thesis.

3.9 Performance Evaluation

3.9.1 Computational Benchmarking

To compare the computational requirements of MediaPipe pose estimation and ball tracking, a benchmarking protocol was implemented. For each method, the following metrics were recorded across 300 consecutive frames from the same video:

- **Processing time**: Wall-clock time per frame in milliseconds

- **CPU utilization**: Process CPU usage as a percentage of one core; values above 100% indicate that several cores were used
- **Memory usage**: Resident set size in megabytes

The two methods perform different tasks on different inputs: MediaPipe processes the full frame, whereas the ball tracker in the analysis pipeline searches only a region of interest (Section 3.3.2). The benchmark therefore indicates the order of magnitude of their computational cost rather than providing a like-for-like comparison.

3.9.2 Benchmark Results

Complete benchmark results are presented in Chapter 4 (Table 4.2). Both methods processed frames faster than the 30 frames per second of typical webcam video on standard consumer hardware without GPU acceleration. Ball tracking was also well above the 60 frames per second of the recordings used in this study, whereas MediaPipe was, on average, slightly below it.

3.10 Data Organization

3.10.1 Output Format

For each recording session, the pipeline writes a CSV file with one row per processed frame in which the sphere was detected (Table 3.3).

The angle columns are filled in the order in which poses were detected and are padded or truncated to the number of sphere samples. Because pose and sphere detections do not occur in exactly the same frames, the angles in a row are not time-aligned with its timestamp. The per-row angles are therefore not used for time-based analysis, and session-level angle statistics (Section 3.6.3) are computed from all detected poses instead.

Table 3.3: Output data structure

Column	Type	Description
timestamp	Float	Time in seconds from the start of the video
x_position	Integer	Sphere x-coordinate in pixels (region of interest, Kalman-filtered)
y_position	Integer	Sphere y-coordinate in pixels (region of interest, Kalman-filtered)
shoulder_angle	Float	Shoulder joint angle (degrees)
elbow_angle	Float	Elbow joint angle (degrees)
wrist_angle	Float	Wrist joint angle (degrees)

3.10.2 Summary Statistics

In addition to the frame-by-frame data, the pipeline writes a JSON report for each session containing the session-level measures described in this chapter: tremor measures (Section 3.7), speed and segment-based bradykinesia indicators (Sections 3.6.2 and 3.8.1), residual-based dyskinesia indicators (Section 3.8.2), joint angle and joint position statistics (Section 3.6.3), and the heuristic screening output (Section 3.8.4). The report also records the calibration details, namely the final calibration factor, the number of estimates, and the standard deviation of the last 100 estimates, together with the tracking-quality measures described below.

3.10.3 Tracking Quality Measures

The quality of sphere tracking in each session is summarised by four measures:

- **Detection rate**: Fraction of processed frames with a valid sphere candidate (Section 3.3.4)
- **Average confidence**: Mean detection confidence (Section 3.3.4) of the detections among the last 30 processed frames
- **Stability**: A score based on the coefficient of variation (CV) of the sphere radius over the last 30 detections, equal to 1 for a CV below 0.15, 0 for a CV above 0.4, and decreasing linearly in between

- **Reliability level**: A label based on the detection rate: high (at least 85%), moderate (at least 70%), low (at least 50%), or insufficient

Because the confidence and stability measures are computed over a sliding window, the values stored in the report describe the end of the session, whereas the detection rate covers the whole session.

3.11 Analysis Pipeline

Figure 3.1 illustrates the complete processing pipeline from video input to the output files.

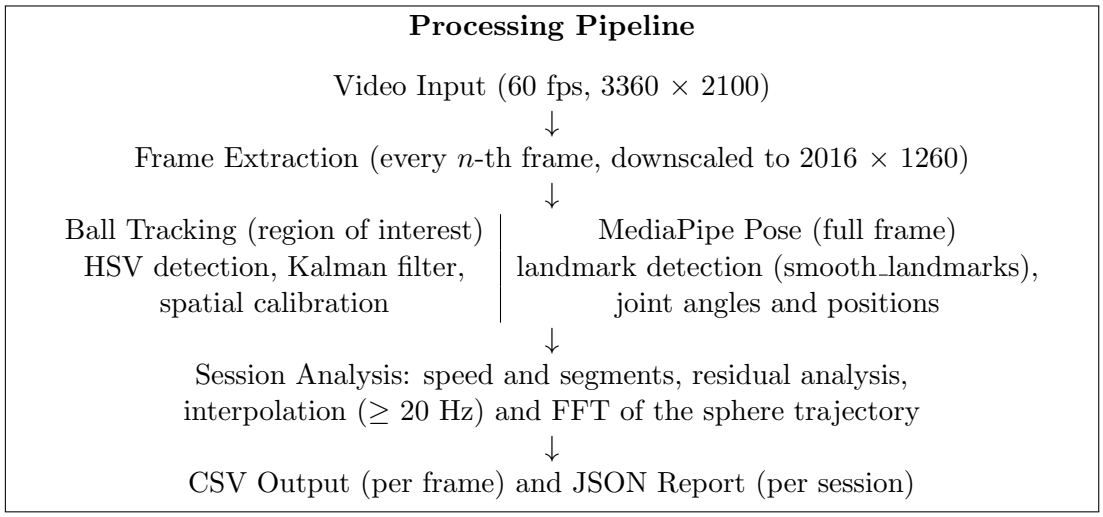


Figure 3.1: Movement analysis processing pipeline. Frames are processed through two parallel tracking pathways; the sphere trajectory is then analysed at session level, with interpolation applied only for the frequency analysis. Per-frame data are written to CSV and session-level measures to a JSON report.

3.12 Statistical Analysis

Given the small patient group (n=4), no inferential statistical tests were performed and no comparisons between groups are reported. Results are summarised descriptively, as

means and standard deviations across sessions.

3.13 Reproducibility

The analysis pipeline, implemented as a single Python script of approximately 2,200 lines, is available upon request. Within the script, spatial calibration, tracking-quality assessment, joint angle analysis, symptom analysis, and sphere tracking are implemented as separate classes, which facilitates adaptation to different recording configurations or tracking targets.

The main run settings (input video, participant identifier, sphere diameter, optional manual calibration, frame-skip, and downscaling factor) are grouped in a configuration block in the main function, while detection thresholds, the region of interest, and filter parameters are defined in the code. Adapting the system to a different site, camera layout, or lighting condition therefore requires editing these values; moving them to an external configuration file would be a natural next step.

Chapter 4

Results and Discussion

This chapter presents the findings from analyzing movement data collected from 50 participants during VRRS rehabilitation sessions.

4.1 Data Collection Summary

4.1.1 Participant Overview

Video recordings from all 51 sessions (50 participants) were successfully processed. Table 4.1 summarises the recording durations.

Table 4.1: Summary of recording durations

Metric	**Volunteers (47 sessions)**	**Patients (4 sessions)**
Recording duration (s)	608.4 ± 77.1	678.9 ± 62.8
Duration range (s)	405.6 – 728.0	619.1 – 766.0

The volunteer group included 47 recording sessions from 46 unique individuals. One participant (subject 27VM/30VM) was recorded twice under different identifiers; both recordings are included. Recording durations averaged approximately 10 minutes, consistent with typical VRRS exercise session lengths.

The patient group consisted of four individuals with diagnosed Parkinson's disease, whose sessions were slightly longer on average than those of the volunteers (Table 4.1).

4.1.2 Detection Rates

Detection success varied between the two tracking methods and across recording conditions. For MediaPipe, the detection rate is the fraction of processed frames in which a pose was detected; for ball tracking, it is the fraction of processed frames with a valid sphere candidate (Section 3.10.3).

MediaPipe pose estimation achieved detection rates ranging from 39% to nearly 100% depending on participant positioning, clothing contrast, and lighting conditions. The lower detection rates occurred primarily when participants moved outside the camera's optimal field of view or when body parts were occluded by furniture or equipment.

Ball tracking achieved detection rates between 20% and nearly 100%, with performance strongly dependent on lighting conditions and the visibility of the blue sphere against the background. Sessions conducted with consistent overhead lighting showed near-perfect detection, while sessions with variable natural lighting or reflective surfaces exhibited reduced detection rates. Eight sessions (seven volunteer sessions and one patient session) had a ball detection rate below 50%, corresponding to an insufficient reliability level (Section 3.10.3); measures derived from the keywand trajectory are least reliable for these sessions.

4.2 Computational Performance

4.2.1 Processing Speed Comparison

A systematic benchmark compared the computational requirements of MediaPipe pose estimation and colour-based ball tracking. Both methods processed the same 300-frame video sequence, with performance metrics recorded for each frame. Table 4.2 presents the results.

Ball tracking was substantially faster than MediaPipe on a per-frame basis, reflecting the computational gap between deep learning-based pose estimation and colour thresholding operations. Because the two methods process different inputs (Section 3.9.1),

Table 4.2: Computational performance comparison (300 frames)

Metric	**MediaPipe**	**Ball Tracking**
Processing time (ms/frame)	17.88 ± 7.78	0.58 ± 0.25
CPU utilization (%)	167.7 ± 26.5	249.2 ± 29.9
Memory usage (MB)	580.4 ± 69.5	489.2 ± 24.2

these values indicate the scale of the difference rather than an exact speed-up factor. The benchmark was also performed on a single video sequence on one testbed configuration (Section 3.3.1), so the reported figures should be read as an illustrative comparison of computational cost rather than as precise, generalisable throughput values.

4.2.2 Resource Utilization

MediaPipe consumed more memory (580 MB versus 489 MB), likely because of the neural network model loaded into memory. CPU utilization was higher for ball tracking (249% versus 168% of one core), likely reflecting OpenCV's use of several cores for colour conversion and morphological operations. Both methods operated well within the capabilities of standard consumer hardware, supporting feasibility for clinical deployment without specialised computing equipment.

4.2.3 Real-Time Capability

Both methods processed frames faster than the 30 frames per second of typical webcam video, with ball tracking providing substantially more processing headroom than MediaPipe. Relative to the 60 fps recordings used in this study, ball tracking kept a large margin, whereas MediaPipe's average processing time was slightly above the 16.7 ms available per frame; real-time MediaPipe analysis of such recordings would therefore require processing every second frame or faster hardware. This margin suggests that ball tracking in particular could accommodate lower-powered hardware or additional real-time processing.

4.3 Position Tracking Results

4.3.1 Spatial Distribution

The blue sphere tracking yielded position data characterising the spatial extent of movements. Positions are expressed in pixels within the tracking region (498 × 375 pixels, Section 3.3.2). Volunteer data showed mean x-position of 173.6 ± 60.5 pixels and mean y-position of 212.6 ± 49.8 pixels, indicating movements centred to the left of and slightly below the centre of the tracking region.

Position variability provides a proxy for movement amplitude. The substantial standard deviations (60.5 and 49.8 pixels in x and y respectively) indicate that participants moved the keywand across a substantial part of the tracking region, consistent with the reaching and pointing tasks characteristic of VRRS exercises. Because these values are in pixels, they are not directly comparable between sessions recorded at different camera distances.

4.3.2 Calibration and Joint Angle Independence

An important methodological point concerns the relationship between spatial calibration and the joint angle measurements described in Section 3.4.3. Joint angles are computed from the geometric relationship between three landmarks, and the calibration factor k (Section 3.3.5) would apply a uniform scaling to their coordinates. A uniform scaling preserves the geometry of the triangle formed by any three landmarks, and therefore preserves the angle at the joint; in the implementation, the angles are in fact computed directly from MediaPipe's normalised coordinates, so the calibration factor never enters the angle computation. For this reason, joint angle results are calibration-independent by construction, and they are not reported as a basis for group comparison in this chapter (see Section 4.4 for the rationale).

The calibration factor is, by contrast, directly relevant to measures derived from the keywand's trajectory, such as displacement, path length, speed, and tremor amplitude,

which depend on absolute scale rather than on relative geometry. In this pipeline, these measures are converted to millimetres at the end of each session (Section 3.5.3), while the position statistics of Section 4.3.1 are reported in pixels.

4.3.3 Calibration Reliability

The initial calibration succeeded within the first 100 frames in 38 of the 51 sessions; in the remaining 13 volunteer sessions it started from the first detection during tracking (Section 3.3.5). Across sessions, the final calibration factor ranged from 0.32 to 1.42 pixels per millimetre (median 0.80), corresponding to an apparent sphere diameter of 16 to 71 pixels (median 40 pixels), which reflects the different camera distances between sessions.

Within sessions, the individual estimates varied considerably: the standard deviation of the last 100 accepted estimates, relative to the final calibration factor, had a median of 22.3% (interquartile range 17.6–26.8%) and exceeded 10% in 48 of the 51 sessions. Part of this variation is detection noise, which the median over 100 estimates suppresses; part reflects genuine changes in the keywand's distance from the camera, which a single scale factor per session cannot capture. Millimetre values derived from the keywand trajectory, such as speed and path length, should therefore be read as approximate.

4.3.4 Tracking Accuracy

To assess the accuracy of ball tracking, the sphere was located manually in 50 frames of session 7VM. The frames were taken at approximately equal intervals across the session from the frames in which the tracker had detected the sphere, excluding the first 20 detections, which are affected by the start-up of the Kalman filter (Section 3.5.3). In each frame, the centre and the edge of the sphere were marked on the full-resolution image without knowledge of the tracker's output, and the marked positions were converted to the coordinates of the tracking region. Distances were converted to millimetres using the size of the sphere marked in the same frame.

In 49 of the 50 frames the tracker followed the sphere; in the remaining frame, at the end of the session, it followed a different blue object. In these 49 frames, the positions stored by the pipeline were a median of 13.5 pixels (about 21 mm) from the marked centre (90th percentile 29.2 pixels, about 47 mm) and lay outside the marked sphere in 16 frames. The colour detections before Kalman filtering were closer, with a median of 9.2 pixels (about 14 mm), and lay outside the marked sphere in only one frame. The additional error of the stored positions was larger when the keywand moved quickly, which is consistent with the Kalman filter lagging behind fast movements at the reduced processing rate of this session (every eighth frame).

The colour detections were also displaced systematically downwards, by a median of 7.4 pixels (Figure 4.1). Because the sphere is a light source, its brightest part appeared almost white in the recordings and fell below the saturation threshold of the colour mask (a median of 31% of the marked sphere was missing from the mask), while the handle directly below the sphere, lit by its blue light, appeared blue and was included in the mask. The detected region was therefore not round (median circularity 0.53), and the radius of its minimum enclosing circle was a median of 1.43 times the radius marked manually. Since the calibration factor is derived from this radius (Section 3.3.5), it was too large by a similar proportion, and millimetre values derived from the keywand trajectory are underestimated by about 30% in this session. These results come from a single session and may differ in sessions with other camera distances or lighting.

4.4 Movement Characterization

Joint angle statistics for the shoulder, elbow, and wrist were computed for all participants following the method described in Section 3.4.3. On inspection, however, none of the three joints showed a difference between the patient and volunteer groups that was large relative to the within-group variability, and, as noted in Section 4.3.2, the angle values themselves are unaffected by spatial calibration and therefore do not benefit from the physical-unit interpretation that motivates reporting other measures in millimetres.

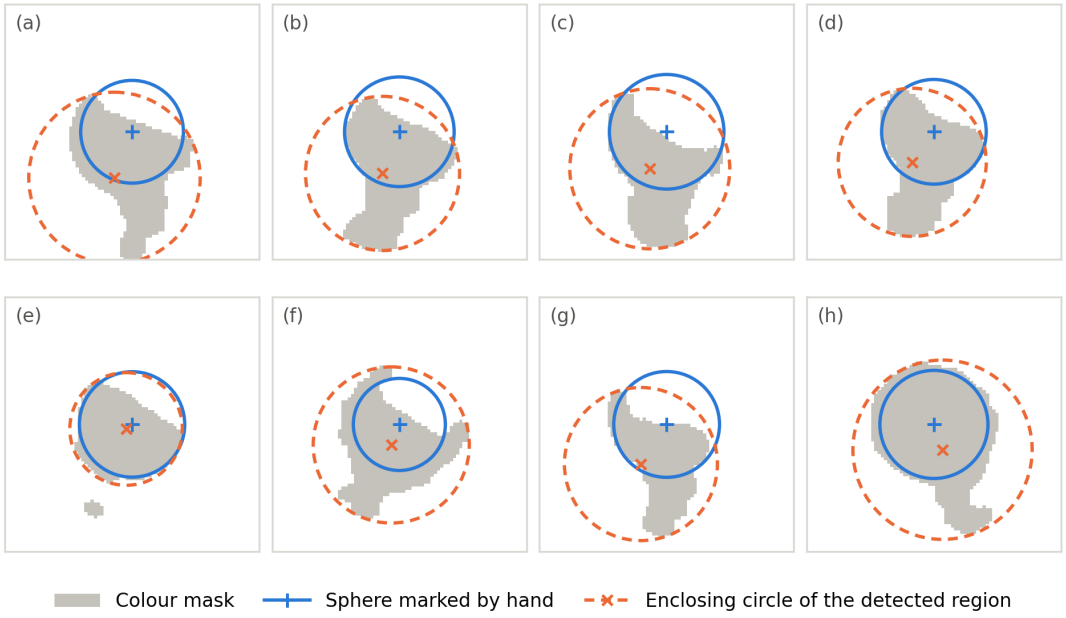


Figure 4.1: Colour mask of the ball tracker (grey) in eight of the 50 checked frames of session 7VM, spread over the session, with the sphere marked by hand (blue, solid) and the minimum enclosing circle of the detected region (orange, dashed). The mask misses the brightest part of the sphere and includes the handle below it. The mask was obtained by applying the colour detection of the pipeline (Section 3.3.2) to the recorded frames.

The angles are also computed from normalised coordinates that are not scaled equally in both directions, so they are relative rather than exact image-plane angles (Section 3.4.3). Given the additional caveat that the patient group is very small (n=4, Section 4.8.2), these joint angle comparisons are not reported here as clinical findings. The angle computation pipeline itself remains part of the system's output and would be a candidate for reassessment with a larger, better-powered patient sample.

The position-based results of this study are therefore the keywand position statistics of Section 4.3.1 and the frequency analysis of Section 4.5. Calibrated keywand measures such as speed and segment amplitude are part of the per-session output (Section 3.10) and are left for analysis in a larger cohort.

4.5 Frequency Analysis

4.5.1 FFT Implementation

Fast Fourier Transform analysis was applied to the sphere trajectory of each session to investigate oscillatory movement components, following the procedure of Section 3.7: positions were interpolated to a uniform rate of at least 20 Hz, band-pass filtered to the tremor band, combined into a magnitude signal, windowed with a Hann function to reduce spectral leakage, and transformed to the frequency domain. Joint angle time series were not analysed in the frequency domain.

4.5.2 Tremor Band Investigation

Tremor-band power, tremor ratio, and dominant frequency were computed for the 3.5–7.5 Hz band corresponding to typical Parkinson's tremor frequencies. As anticipated in Section 3.7.3, this analysis yielded inconclusive results for tremor detection: no consistent peaks appeared in the 4–6 Hz tremor band.

Although the band-pass filter removes most of the slow voluntary movement before the spectrum is computed, active VRRS exercise still produces energy in the tremor band, for example through rapid direction changes, and tracking artefacts such as detection gaps bridged by interpolation or jumps between detected candidates can add further content. Tremor-band power measured during the exercise therefore cannot be attributed to tremor. Moreover, the native sampling rate of the sphere positions ranged from 4.9 to 14.5 Hz across sessions, below the 15 Hz needed to cover the whole band: in 25 of the 51 sessions the band lay entirely above the Nyquist frequency, and in the remaining 26 it was only partly covered (Section 3.5.2).

4.5.3 Implications for Tremor Assessment

These results confirm the methodological limitation identified in Section 3.7.3: rest tremor cannot be meaningfully assessed during active, goal-directed movement. Valid

tremor assessment would require dedicated rest periods, IMU sensors, or exercise designs that include static holding tasks. The FFT infrastructure developed here remains applicable given appropriate data collection protocols.

4.6 Data Quality Assessment

Figure 4.2 shows the tracking-quality measures of Section 3.10.3 for session 7VM of ball tracking. The session reached a detection rate of 78.5%, which corresponds to a moderate reliability level. Its stability score of 0.65 indicates that the apparent size of the sphere varied moderately towards the end of the session. Its average confidence of 0.28 should be read with care: the area term of the confidence (Section 3.3.4) reaches its maximum only for blobs of 10,000 pixels, much larger than the sphere typically appears in these recordings, so the confidence is dominated by circularity and remains well below 1 even for correct detections.

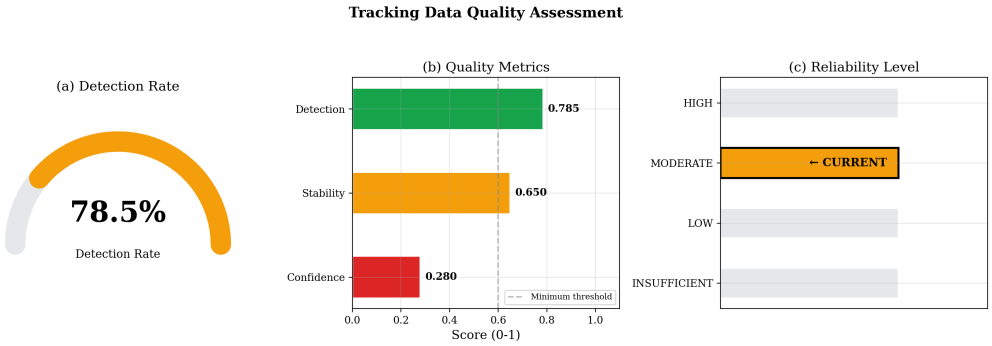


Figure 4.2: Tracking-quality measures of ball tracking for session 7VM: (a) detection rate, (b) detection rate, stability, and average confidence, and (c) reliability level.

4.7 Comparison of Tracking Methods

4.7.1 Complementary Strengths

The two tracking approaches offer complementary capabilities. Table 4.3 summarizes their relative strengths.

Table 4.3: Comparison of tracking method characteristics

Characteristic	**MediaPipe**	**Ball Tracking**
Processing speed	Moderate	Substantially faster
Memory requirement	Higher (580 MB)	Lower (489 MB)
Output data	Body landmarks, angles	Keywand position and size
Robustness to lighting	Moderate	Low
Requires specific equipment	No	Blue sphere
Anatomical insight	Yes	No

MediaPipe provides richer data through body landmark detection, enabling joint angle computation and anatomical analysis. Ball tracking offers superior speed and lower computational requirements but provides only the position and apparent size of the tracked object.

4.7.2 Combined Approach Potential

A hybrid system could leverage both methods: ball tracking for continuous high-speed position monitoring and MediaPipe for periodic pose snapshots providing anatomical context. Such an approach would balance computational efficiency against data richness, potentially enabling real-time feedback during exercises while capturing clinically relevant movement parameters.

4.8 Discussion

4.8.1 Principal Findings

The results presented in Sections 4.1 through 4.7 establish the technical feasibility of augmenting VRRS rehabilitation sessions with markerless motion capture. Both tracking methods processed frames faster than typical 30 fps webcam video on consumer hardware (Table 4.2), and the system successfully processed the recordings of all 50 participants (51 sessions).

The most methodologically important observation from this study concerns the relationship between spatial calibration and the two families of features it can produce. Position- and displacement-derived measures benefit directly from calibration, since they depend on absolute physical scale; joint angle measures, by construction, do not, since they depend only on the relative geometry between landmarks (Section 4.3.2). This distinction shapes which measures are meaningful to report as calibrated, physical-unit findings, and which are better understood as scale-invariant descriptors of posture. Given this, and given that no joint showed a between-group difference large relative to within-group variability in a sample where the patient group numbers only four individuals, joint angle comparisons are not presented here as clinical findings; they remain, however, part of the system's technical output and a candidate for re-examination with a larger patient cohort. Finally, the frequency analysis showed that tremor cannot be assessed from the keywand trajectory during active VRRS exercise (Section 4.5).

4.8.2 Limitations

Several limitations constrain interpretation of these findings:

Sample size. The patient group (n=4) is too small for statistical comparison with the volunteer group (n=46). Any preliminary observations from this cohort should be treated as hypothesis-generating rather than conclusive, and would require validation in larger cohorts.

Clinical correlation. This study did not collect clinical assessments (e.g., MDS-UPDRS scores) that would enable correlation between extracted movement features and established severity measures.

Single-camera limitations. Monocular video provides limited depth information, potentially affecting accuracy of movements perpendicular to the camera plane. Multi-camera setups or depth sensors could address this limitation.

Naturalistic recording conditions. As described in Section 3.2.2, recordings were deliberately collected under varying, uncontrolled conditions (camera distance, lighting, background) to reflect realistic deployment scenarios rather than a standardised laboratory setup. While this strengthens the ecological validity of the findings, it also introduces session-to-session variability that likely contributes to the range observed in detection rates (Section 4.1.2) and may affect the consistency of derived measurements across sessions.

Implementation details. Several properties of the implementation, described in Chapter 3, limit the precision of the derived measures: the start-up transient of the Kalman filter inflates the peak speed of each session (Section 3.5.3); joint angles are computed from coordinates that are not scaled equally in both directions (Section 3.4.3); the joint angles stored per row in the CSV output are not time-aligned with the sphere samples (Section 3.10.1); the region of interest and colour thresholds had to be adjusted by hand for some recordings, so not all sessions were processed with identical settings (Section 3.3.2); and the frame-skip setting was not the same for all recordings, so the effective sampling rate differs between sessions (Section 3.5.1). In addition, the accuracy check showed that the colour mask misses the overexposed part of the light sphere and includes the handle below it, which displaces the detected positions downwards and makes millimetre values too small (Section 4.3.4).

4.8.3 Relation to Prior Work

The findings align with previous validation studies of MediaPipe for clinical movement analysis. Stenum et al. showed that gait parameters measured with video-based pose

estimation agree with three-dimensional motion capture in unimpaired adults and in people after stroke or with Parkinson's disease [23]. The present study applies MediaPipe to upper limb rehabilitation, a different clinical context, although it did not include a reference measurement system with which to assess accuracy.

The ball tracking performance is consistent with the general expectation that classical colour-thresholding methods substantially outperform deep learning-based pose estimation in raw throughput, particularly for a favourable, uniformly coloured target such as the keywand sphere.

4.8.4 Future Directions

Several extensions could enhance the clinical utility of this approach. Collecting paired clinical assessments (MDS-UPDRS scores) would enable correlation of video-based features with established severity measures. Longitudinal tracking through repeated assessments could reveal sensitivity to disease progression or treatment effects. Integration with the VRRS display could provide real-time movement quality feedback. Validation in home environments would support remote rehabilitation monitoring. Including patients with other neurological conditions could identify condition-specific movement signatures. A larger patient cohort would also allow the joint angle measures described in Section 4.4 to be properly re-examined with adequate statistical power. On the technical side, initialising the Kalman filter at the first detection, applying the calibration before filtering, aligning joint angles with the sphere samples in the output, and moving parameters to a configuration file would address the implementation limitations noted in Section 4.8.2.

Chapter 5

Conclusion and Future Works

This thesis investigated the feasibility of augmenting Virtual Reality Rehabilitation System (VRRS) platforms with markerless motion capture, using computer vision to extract clinically relevant movement data from standard video recordings without requiring additional specialised sensors.

5.1 Summary of Approach and Results

The study combined two complementary tracking methods: MediaPipe pose estimation for anatomical landmark detection and joint angle computation, and colour-based tracking of the VRRS blue sphere keywand in the HSV colour space. Data from 50 participants (46 healthy volunteers, 4 Parkinson's patients), recorded in 51 sessions, were processed through a pipeline incorporating Kalman filtering, spatial calibration, and frequency analysis with temporal interpolation.

The key findings, detailed in Chapter 4, can be summarised as follows. Both tracking methods processed frames faster than typical 30 fps webcam video on consumer hardware, with ball tracking substantially faster than MediaPipe on a per-frame basis (Table 4.2). Spatial calibration was found to be relevant specifically to position- and displacement-derived measures, such as the displacement, speed, and tremor amplitude of the keywand, while joint angle measures were shown to be calibration-independent

by construction, since a uniform scaling of coordinates does not alter the geometry of the triangle formed by three landmarks (Section 4.3.2). This distinction informed which measures are meaningful to report in physical units. FFT-based tremor detection proved unsuitable for the active movement context of VRRS exercises, as voluntary movement and tracking artefacts also produce energy in the tremor band and mask any underlying tremor components (Section 4.5).

5.2 Contributions

This thesis makes five contributions to the intersection of computer vision and neurological rehabilitation:

Feasibility demonstration. Meaningful movement data can be extracted from standard VRRS video recordings using free, open-source tools without additional sensors or equipment modifications. The system processed all 50 participants successfully.

Performance benchmarking. The benchmark comparison of pose estimation versus colour-based tracking provides practical guidance for system design. Ball tracking suits embedded or resource-constrained applications, while MediaPipe supports detailed anatomical analysis.

Clarification of calibration scope. The analysis identified and formalised an important methodological distinction: spatial calibration meaningfully affects position- and displacement-based measures but does not affect angle-based measures, since angles are invariant to uniform coordinate scaling. This distinction is directly relevant to any future markerless system that combines both position and angular measurements, and clarifies which outputs of such a system require physical-unit calibration to be clinically meaningful.

Context-specific limitation identification. The failure of tremor detection during active movement illustrates the importance of matching analysis methods to appropriate data collection protocols, preventing future researchers from repeating this approach without protocol modifications.

Open methodology. The processing pipeline, comprising approximately 2,200 lines of documented Python code organised into separate classes for calibration, tracking, and analysis, provides a foundation for adaptation across different clinical sites and recording configurations.

5.3 Limitations

Beyond the limitations discussed in Section 4.8.2, two additional constraints merit emphasis at the thesis level:

Cross-sectional design. The study captured single sessions from each participant rather than repeated assessments over time. Longitudinal data would be required to evaluate sensitivity to disease progression or treatment effects, which are among the most clinically valuable applications of movement monitoring [24].

Absence of clinical correlation. No standardised clinical assessments (e.g., MDS-UPDRS scores) were collected alongside video recordings. Without paired clinical data, the relationship between extracted movement features and established severity measures remains unknown. This is the single most important gap to address in future work [16].

The remaining limitations (sample size imbalance, single-camera geometry, naturalistic recording conditions, and implementation details) are discussed in Section 4.8.2, and the active movement context in Section 4.5.

5.4 Future Directions

The most impactful next steps, in order of priority:

Expanded clinical validation. Recruiting a larger patient cohort with paired MDS-UPDRS assessments would transform this from a feasibility study into a clinical validation study, and would also allow the joint angle measures excluded from this study's clinical comparisons (Section 4.4) to be re-examined with adequate statistical power. Multi-site recruitment would enhance generalisability.

Longitudinal monitoring. Repeated assessments within individuals over time could reveal sensitivity to disease progression, medication effects, or rehabilitation outcomes, supporting the use of video-based metrics as outcome measures in clinical trials or routine care [25].

Protocol modifications. Incorporating brief rest periods (10–15 seconds) into VRRS sessions would enable tremor assessment using the existing video infrastructure. Specific movement tasks designed to elicit bradykinesia could expand the range of assessable symptoms.

Home deployment. Validating system performance in home environments would support telerehabilitation applications. This is particularly relevant given increasing interest in decentralised care and remote patient monitoring [26].

Real-time VRRS integration. Connecting the analysis pipeline to the VRRS display could provide patients immediate movement quality feedback, potentially enhancing therapeutic engagement. Ball tracking's speed makes this practical for real-time applications.

Multi-condition comparison. Including patients with essential tremor, stroke, or multiple sclerosis could identify condition-specific movement signatures. The VRRS platform already serves diverse neurological populations, providing natural opportunities for such comparisons [2].

5.5 Concluding Remarks

Accurate motion capture was historically available only in specialised laboratories with expensive marker-based systems. Deep learning-based pose estimation has changed this, making quantitative movement assessment accessible through commodity hardware [3, 23]. This thesis demonstrates that accessibility in the context of neurological rehabilitation, showing that standard cameras and open-source software can extract meaningful movement data from VRRS sessions.

The path from technical feasibility to clinical impact requires validation in larger co-

horts, correlation with established clinical measures, and demonstration of sensitivity to meaningful change. The foundations established here, including the processing pipeline, performance benchmarks, and the clarified role of spatial calibration across different measurement types, provide a starting point for that work.

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